Block to Python Bridge

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected paper-to-Python record.

Explain the mapping, predict packing output, repair a misconception and test a boundary.

Keep one habit: Keep code version, input assumptions, intermediate state, output status and support used together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Requests Responses and JSON

Conversation State and Memory

Model Memory vs Retrieval

W1 · Explain the paper-to-code mapping

> Name inputs, assignment, multiplication and display.

Your evidence record

- Packing W2: packs 4, per_pack 6 gives 24; changing packs to 5 gives 30. Assignment stores a value; print displays it. Book-count procedure: start count 0 before loop; add 1 only when pages<100; [80,150,90]→2, [80,150,100]→1, empty list→0. The numeric-list function assumes supplied nonnegative integer page counts; a missing record field requires a separate guard, not invented zero.

- Write the complete four-line script for 4 packs of 6. Beside each line explain what it does. Which line displays information?

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Predict then observe packing outputs

> Keep predictions, traces and actual runs distinguishable.

Your evidence record

- Packing W2: packs 4, per_pack 6 gives 24; changing packs to 5 gives 30. Assignment stores a value; print displays it. Book-count procedure: start count 0 before loop; add 1 only when pages<100; [80,150,90]→2, [80,150,100]→1, empty list→0. The numeric-list function assumes supplied nonnegative integer page counts; a missing record field requires a separate guard, not invented zero.

- Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual **output**, any error and what changed. If working on paper, label both results as traces.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Audit assignment versus display

> An equals sign binds a name; print displays the value.

Your evidence record

- Packing W2: packs 4, per_pack 6 gives 24; changing packs to 5 gives 30. Assignment stores a value; print displays it. Book-count procedure: start count 0 before loop; add 1 only when pages<100; [80,150,90]→2, [80,150,100]→1, empty list→0. The numeric-list function assumes supplied nonnegative integer page counts; a missing record field requires a separate guard, not invented zero.

- Move print(total) to the first line of a fresh script. Predict what happens and why. Put the lines back into an order that meets the requirement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a revealing translation case

> Predict a boundary before running and preserve actual evidence.

Your evidence record

- Packing W2: packs 4, per_pack 6 gives 24; changing packs to 5 gives 30. Assignment stores a value; print displays it. Book-count procedure: start count 0 before loop; add 1 only when pages<100; [80,150,90]→2, [80,150,100]→1, empty list→0. The numeric-list function assumes supplied nonnegative integer page counts; a missing record field requires a separate guard, not invented zero.

- Explain the difference between a successful run and a correct calculation. State one limit of this **program** and what you would keep so another person can check your work.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Trace the new sum/reset procedure independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Trace both versions of the supplied sum procedure in order, keeping each intermediate total. Compare initialization before the loop with initialization inside it. This fresh task sums values; do not copy the book counter's increment-by-one rule.

Set total=0, then for values2,4,9 perform total=total+value. Trace the result. What changes if total=0 moves inside the loop before that addition?

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Implement small transparent **function**s for selection, validation, and citation formatting.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Predict then observe for From a paper **algorithm** to Python. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for From a paper **algorithm** to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Write Python that totals [3,5,7], then trace each total. Explain how indenting 'total=0' inside the loop changes the result.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A block **program** sets total=0 and adds each value in [2,4,6].

Translate and trace.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Trace this Python: score = 2; then if score >= 2: score = score + 3. What is the final value, and what changes if the **condition** becomes score > 2?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 23 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Write Python that totals [3,5,7], then trace each total. Explain how indenting 'total=0' inside the loop changes the result.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Additional practice · Earlier packing transfer

REFERENCE · SECTION 15

Additional practice · Earlier packing transfer

> This retained task differs from the selected sum/reset assessment.

Earlier task and saved response

- A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using trays, per_tray and tiles. Predict its **output**. Then change only trays to 0 and predict again. Explain why adding the two quantities is wrong.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

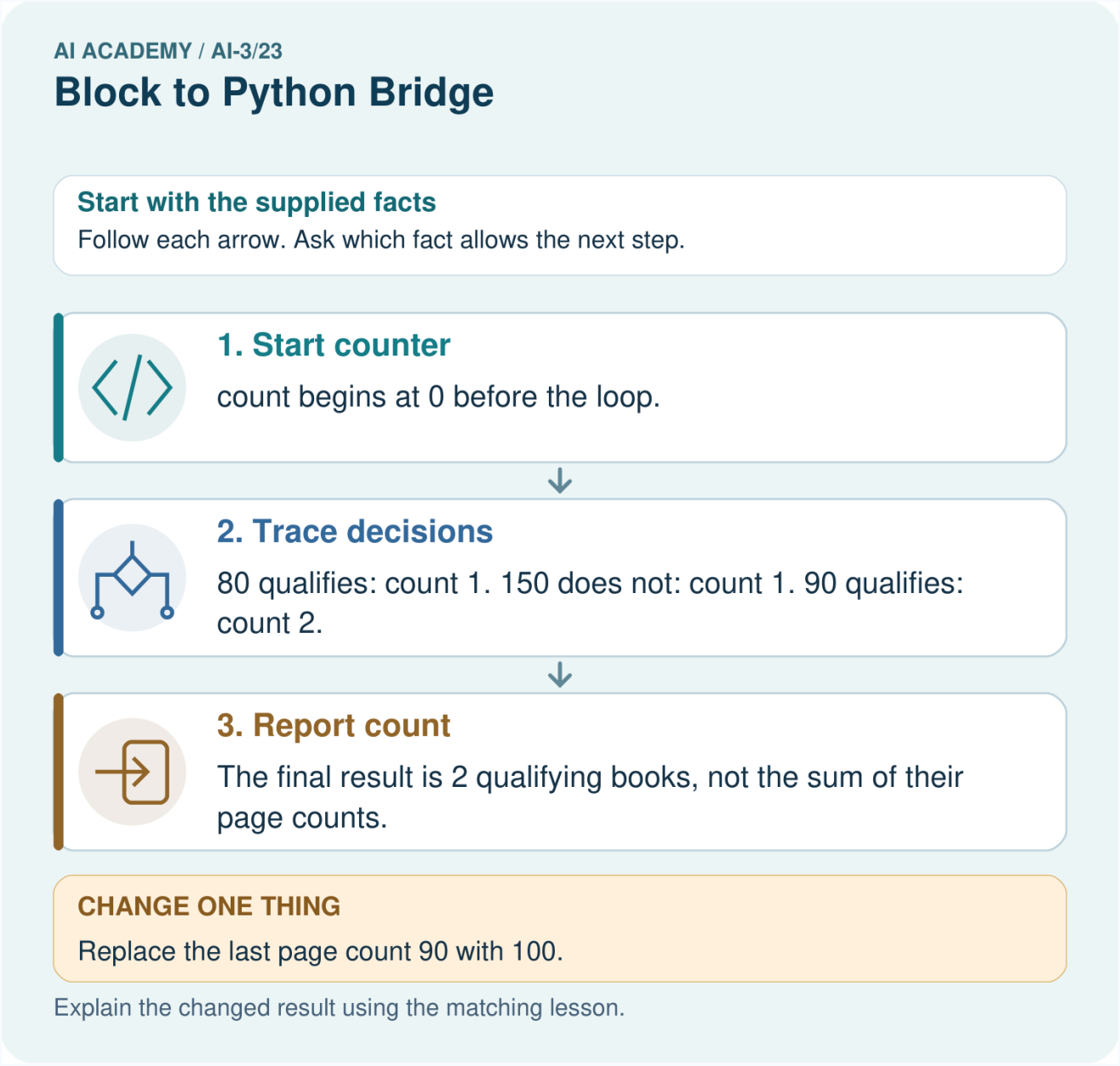
> Keep the explanation and the support you used.

Use this when helpful

Translate a “count qualifying books” paper procedure into a Python-style trace.

Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the changed **list** and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My picture-based explanation and changed-case prediction**

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week’s record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Translate a “count qualifying books” paper procedure into a Python-style trace.

- Which labelled step matches this detail? The final result is 2 qualifying books, not the sum of their page counts.
- 1. Trace decisions
- 2. Start counter
- 3. Report count
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Report count The final result is 2 qualifying books, not the sum of their page counts. Read the picture in order.

  The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the first explanation

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: Replace the last page count 90 with 100. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The final count becomes 1.
- Hint if needed: Use the changed **condition** and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: The final count becomes 1. The comparison is strictly less than 100, so the boundary value does not qualify.

My selected conclusion, deciding fact and explanation

▼ Check the changed case

Repair a missing building block

Repair and save your work

Review Requests Responses and JSON (ai-3/22). Goal: Parse JSON into Python records.
Short practice: Would converting the text silently prove the source count is correct?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Conversation State and Memory (ai-3/20). Goal: Explain the **mechanism** and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Model Memory vs Retrieval (ai-3/17). Goal: Explain the **mechanism** and terminology of retrieval and indexing. Short practice: Why can the assistant not certify 15: 00 when the notice is missing?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Block to Python Bridge: the case

AI ACADEMY / AI-3/23 · 01 · LOOK AT THE FACTS

Block to Python Bridge: the case

The situation

Translate a “count qualifying books” paper procedure into a Python-style trace.

Supplied evidence

Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain block to python bridge using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

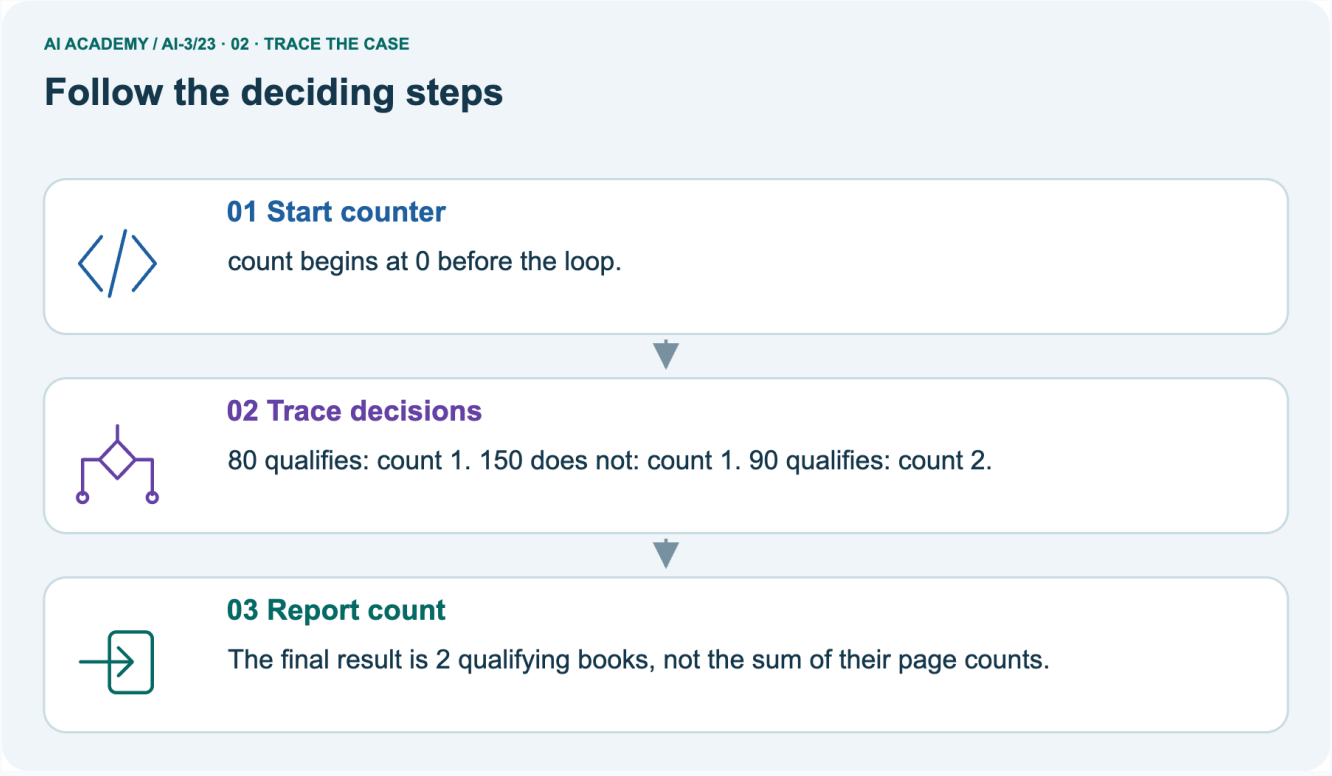
Translate a “count qualifying books” paper procedure into a Python-style trace. Supplied facts: Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

> **My picture explanation, deciding evidence and one limit**

02. Follow the deciding steps



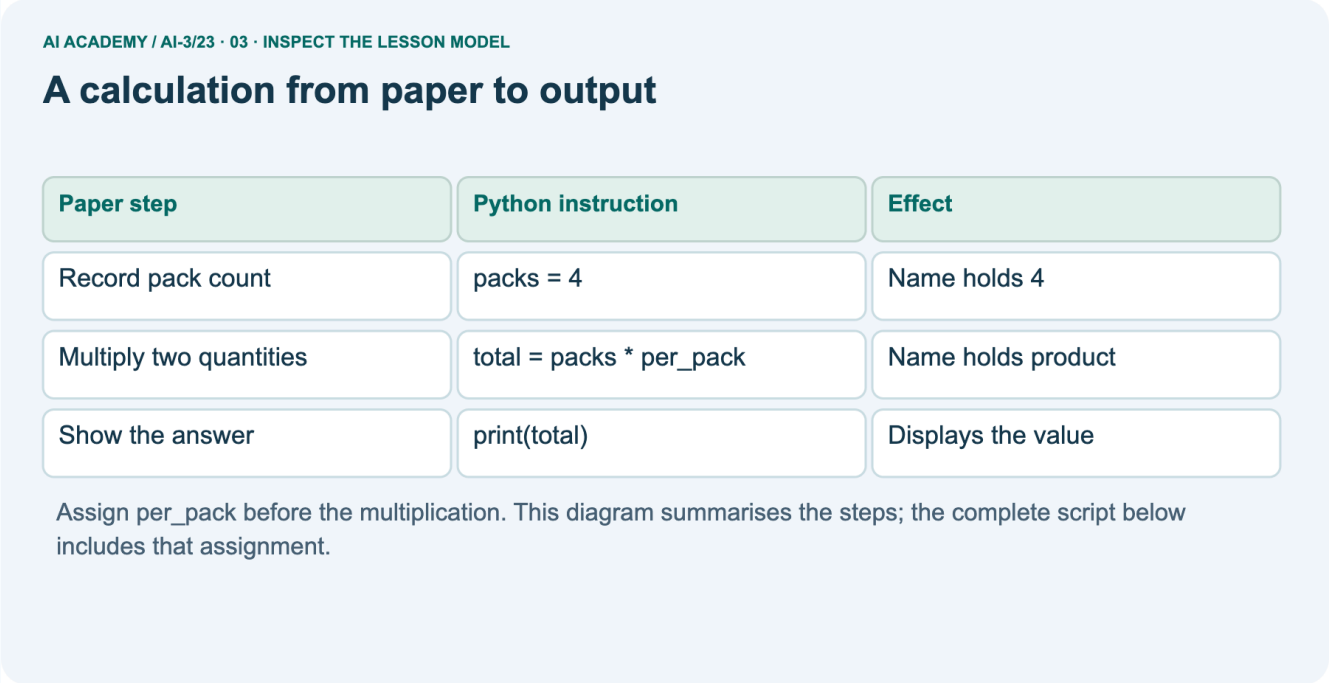
Follow the labelled picture with the matching explanation. Open the vector diagram

Start counter: count begins at 0 before the loop. → Trace decisions: 80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2. → Report count: The final result is 2 qualifying books, not the sum of their page counts.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A calculation from paper to output



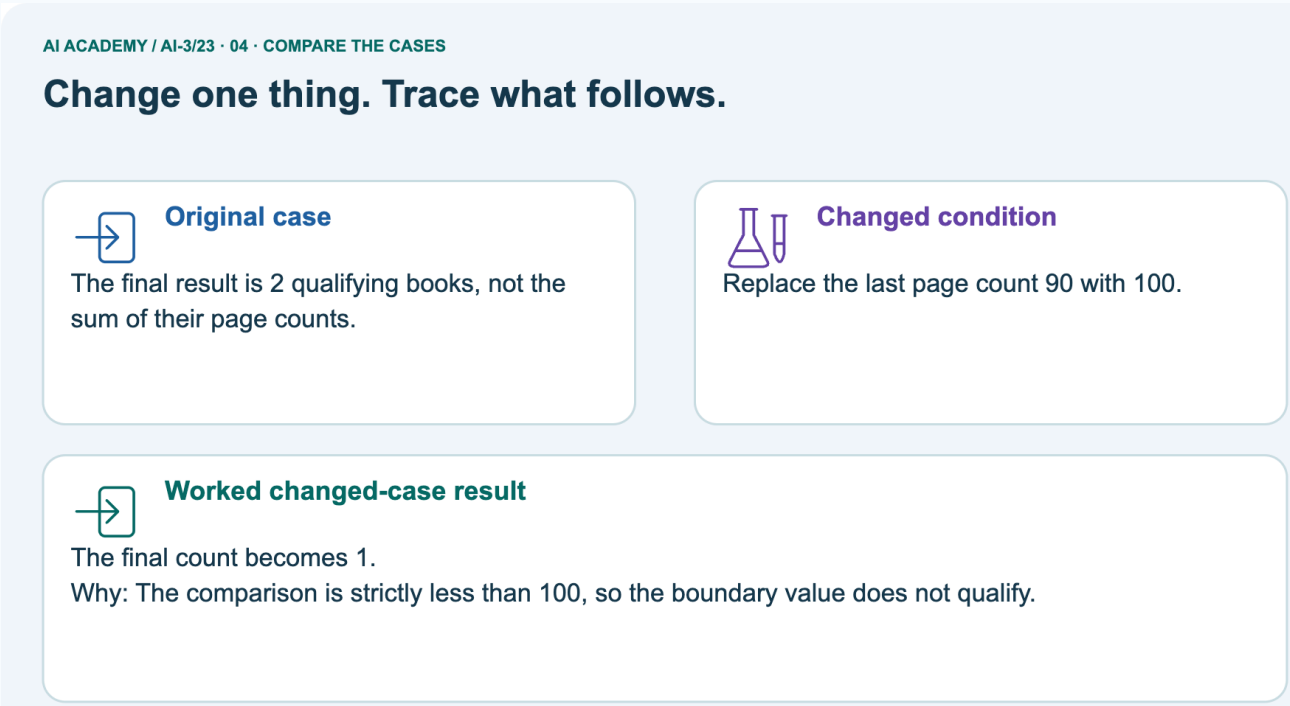
Follow the labelled picture with the matching explanation. Open the vector diagram

A calculation from paper to **output**. Record pack count | packs = 4 | Name holds 4. Multiply two quantities | total = packs * per_pack | Name holds product. Show the answer | print(total) | Displays the value

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

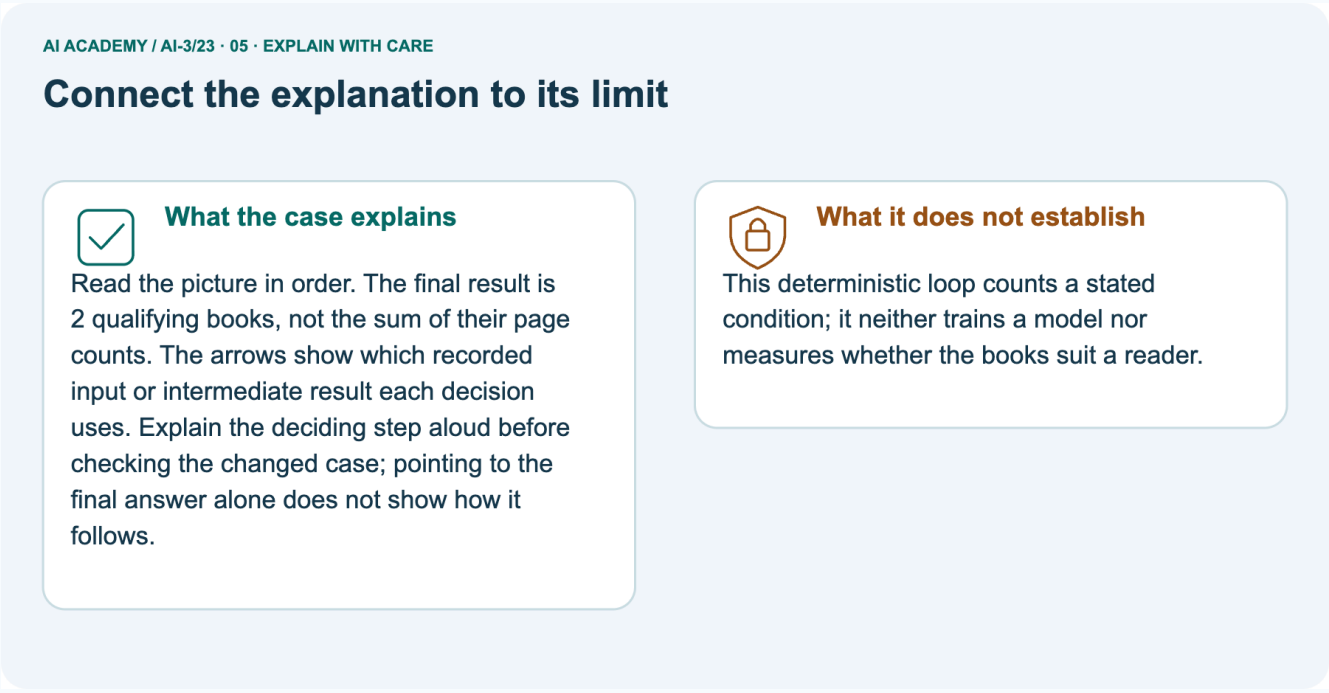
Original case: The final result is 2 qualifying books, not the sum of their page counts. Changed **condition**: Replace the last page count 90 with 100. Expected result: The final count becomes 1.

Reason: The comparison is strictly less than 100, so the boundary value does not qualify.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded **input** or intermediate result each decision uses.

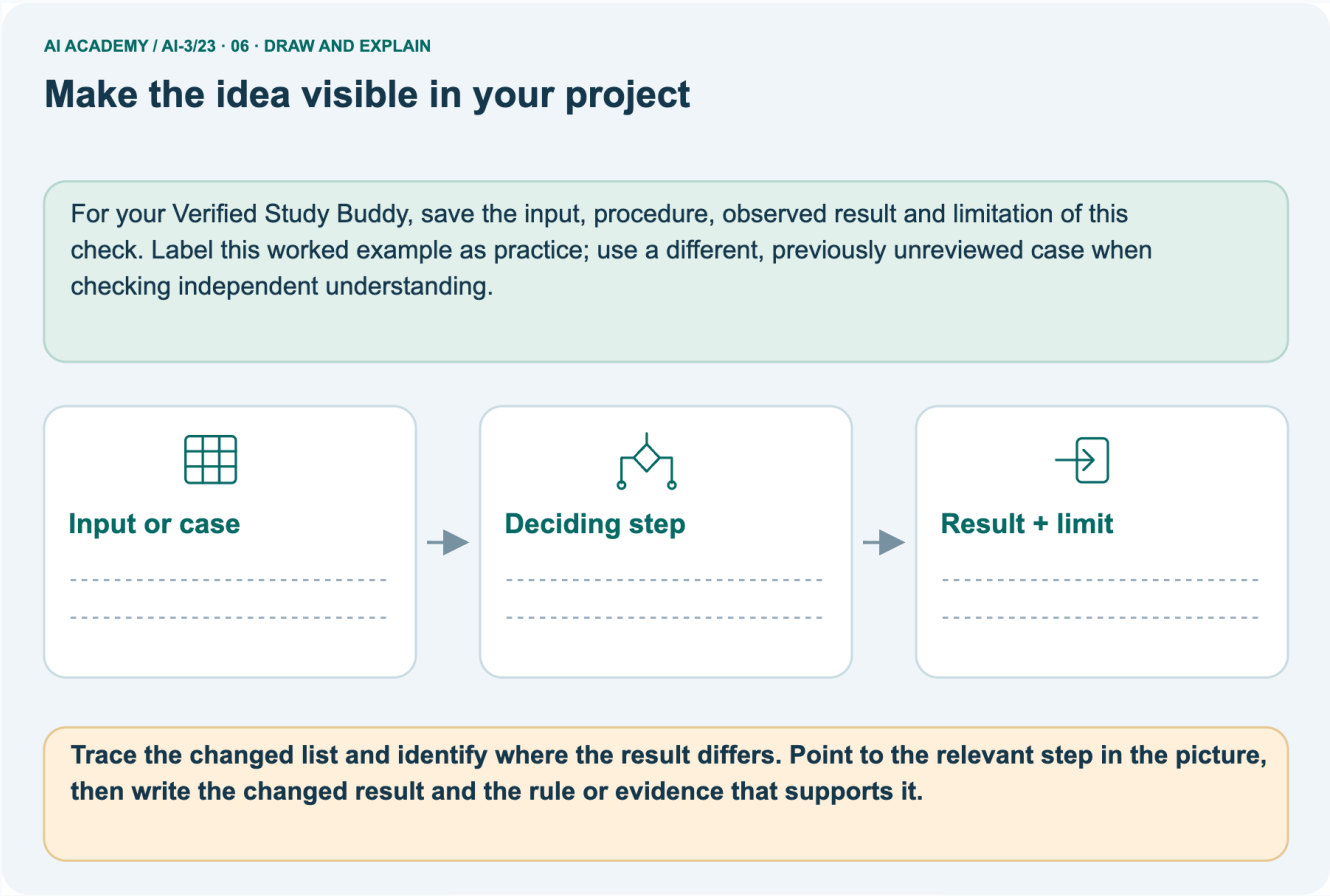
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: This deterministic loop counts a stated **condition**; it neither trains a model nor measures whether the books suit a reader.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Trace the changed **list** and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Earlier wording, alternatives and picture transcripts**

01. Block to Python Bridge: the case

02. Follow the deciding steps

03. A calculation from paper to output

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.